



Transformer with token attention and attribute prediction for image captioning

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ARTICLE INFO

Editor: Silvio Barra

MSC:

41A05

41A10

65D05

65D17

Keywords:

Image Captioning

Token Attention

Vision Transformers

ABSTRACT

Recently, Vision Transformers (ViTs) have become the mainstream models in image captioning tasks. ViTs take all image tokens as inputs to extract visual features, which may cause concerns about worthless tokens, and meanwhile lead to a huge amount of computation. This paper proposes a novel token reduction module to remedy this drawback. Specifically, the module employs ViTs to embed the input tokens, and adaptively learns informative visual tokens in way of token attention on the channel-spatial granularity. Furthermore, an attribute prediction module is designed to strengthen the relationship between vision and language. Technically, the attribute prediction is achieved via a classifier in form of Multi-Layer Perceptron (MLP). Both the visual representations and attribute representations are obtained by Transformers, which are then combined as the input of the Transformer decoder for caption generation. All of the modules are constructed in an encoder–decoder framework and support the end-to-end learning. Experiment results have shown that our approach can effectively reduce the computational cost of ViTs while maintaining comparable performance on the MS COCO and NoCaps datasets. For example, by pruning more than 70% of the input tokens, our approach greatly reduces GFLOPs by 41% ~ 47%, while preserving its accuracy of a 142.1 CIDEr score on the MS COCO dataset.

1. Introduction

Image captioning [1,2] aims to describe images with natural language. It is a challenging task that requires both scenery comprehension as well as language understanding. Various captioning models [3,4] have been developed within the encoder–decoder frameworks for this issue. The encoders are responsible for visual feature extraction and decoders produce the truthful captions. The goodness of the models is largely determined by the learned representations, whether they are discriminative enough to reduce the gap between the visual contents and the semantic description sentences is important.

Recent developments on Transformers [5] have shown the state-of-the-art performances in natural language processing. Its success is largely due to its ability in modeling long-range dependencies. Nowadays, its usage has been extended to vision-language tasks, like text-to-image generation [6], VQA [7], image captioning [8], etc. Among them, Visual Transformer (ViT) [9] is the seminal framework. The advantage of ViTs is that a global representation of the input image can be captured at the beginning of visual perception, which can replace

the convolutional neural networks that obtain image features layer by layer in deep structures.

However, ViTs need to compute all patches (tokens) when processing images. Given an image with 512×512 pixels, for example, suppose that it is divided into 1024 tokens, then the standard ViTs will implement a total of over 4.6 Giga floating point operations (GFLOPs). That is, the Transformer with dense tokens will cause heavy memory cost. In addition, it could involve worthless tokens in computation for image captioning, especially including those cluttered backgrounds. Considering that the computation in the Transformer increases nonlinearly with the number of tokens, an effective way is to prune tokens of less importance from the input sequence of patches.

Furthermore, semantic attributes play equally important roles in image captioning tasks as visual features. Due to the semantic gap, practice indicates that not all of the generated words correspond well to image contents, especially for the words describing complex relationships between objects. However, attributes can directly guide the sentence generation without considering the cross-modal gap between vision and language.

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<https://doi.org/10.1016/j.patrec.2024.11.022>

Received 20 October 2023; Received in revised form 26 October 2024; Accepted 19 November 2024

Available online 26 November 2024

0167-8655/© 2024 Published by Elsevier B.V.

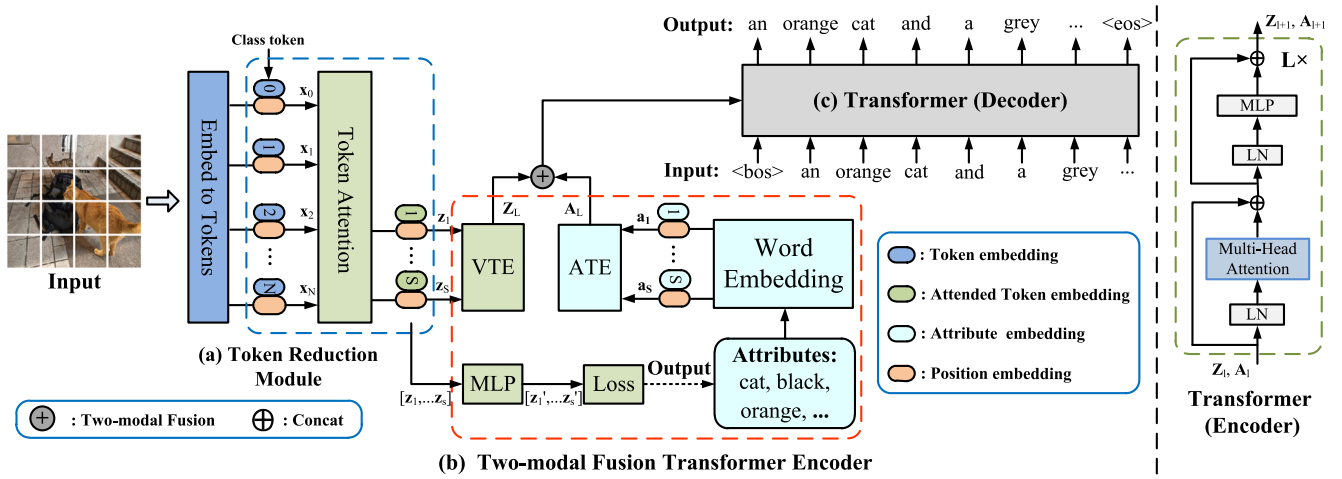


Fig. 1. The overview of the proposed framework. Our model has three modules: (a) The module in the blue dotted box is our proposed attention based token reduction module, which aims to eliminate redundant tokens. (b) The module in the red dotted box is our proposed two-modal fusion Transformer encoder, which consists of VTE (Visual Token Encoder) and ATE (Attribute Token Encoder). (c) The module in the grey box is the Transformer decoder, which can generate corresponding captions. In addition, the module in the green dotted box shows the details of one layer in the Transformer both for vision and attribute, where LN stands for Layer Normalization.

Motivated by the above observations, this paper develops the conception framework of *token attention* to generate a small number of representative tokens for image captioning tasks. That is, instead of employing all of the tokens formatted by uniform splitting, token attention selects to project them non-linearly into a feature space to reduce the redundant components. In addition, a two-modal fused Transformer with the encoder–decoder architecture is constructed to enhance the performance. The encoder learns the visual and attribute representations from the learned tokens, while the decoder generates the sentence based on the fusion features of the two representations.

Fig. 1 shows the overview of the proposed model. The contributions of this paper are summarized as follows:

- Token Reduction Module (TRM) is proposed to generate a smaller number of tokens for image captioning, which greatly reduces ViTs' computation. This module pays more attention to the foreground objects and less attention to the ambiguous backgrounds, and mainly operated by the channel-spatial attention mechanism.
- A two-modal fusion Transformer is developed for vision and attribute encoders. Its visual sub-module is to learn the visual representations, which are further reused to predict the attributes of the image. In this process, attributes are obtained via a classifier in form of Multi-Layer Perceptron (MLP). In parallel, its attribute sub-module is to learn semantic representations from the predicted attributes. Both visual and attribute representations are fused together for image captioning.
- Extensive comparative experiments and ablation study show that our approach achieves competitive performances on the MS COCO and NoCaps datasets, compared with the state-of-the-art models.

2. Related work

2.1. Image captioning with attention mechanisms

In recent years, a lot of efforts have been made under the encoder–decoder frameworks [10] for image captioning. Among the existing approaches, attention mechanisms have been widely used to enhance the performance. For example, bottom-up and top-down attention mechanisms [11] have been proposed to encode the fine-grained visual details. In this framework, bottom-up attention is developed to extract informative regions of the detected objects, and top-down attention is

employed to associate those relevant regions with the words. Then, in Wang et al. [12], two Visual Persistence modules were built in encoder and decoder. Respectively, the two modules perform the task of exploring object areas and evaluating the correlation between current and previous attention outputs. Later, the Retrospect Network for Image Captioning (RNIC) model was developed in Yi et al. [13], which introduces the textual attention that helps the model learn both visual and textual features, and can capture captions with minimal information.

Some recent approaches have explored the use of Transformers [5] in vision-language tasks. Typically, Zhou et al. [14] introduced a model of Compact Bidirectional Transformer for Image Captioning (CBTIC), which can leverage bidirectional context implicitly and explicitly to guarantee the final caption chosen from either left-to-right or right-to-left flow. Later, Nguyen et al. [15] proposed a Transformer-only architecture, dubbed Grid and Region based Image captioning Transformer (GRIT), which utilizes two visual features to generate better captions. Recently, Lim and Chan [10] proposed a Mask Captioning Network (MaC), which comprises an object layer and a background layer to capture objects and scenes of an image to generate a sentence.

However, the aforementioned approaches need to process a large number of densely sampled token to extract image features. Dealing directly with these inputs will bring the irrelevant visual patches for captioning and thus cause a huge amount of computations for training and prediction.

2.2. Tokens reduction

Visual Transformers (ViTs) [9] require a huge computational cost, due to the long input length. In order to reduce the number of tokens, some works select to solve this issue via image classification tasks. Along this line, Rao et al. [16] proposed the dynamic token sparsity to prune progressively the redundant tokens. Later, Ryoo et al. [17] introduced to mine the important tokens in visual data, indicating that very few tokens may be sufficient for visual understanding. Meng et al. [18] developed an end-to-end Adaptive Transformer (AdaViT), which can determine the usage of the patches, heads and layers of ViTs.

In summary, seldom works have made attempts to reduce tokens for image captioning. This motivated us to develop the *token learner* under the ViTs framework, in order to identify the important tokens and ignore the irrelevant ones for sentence generation and computation reduction.

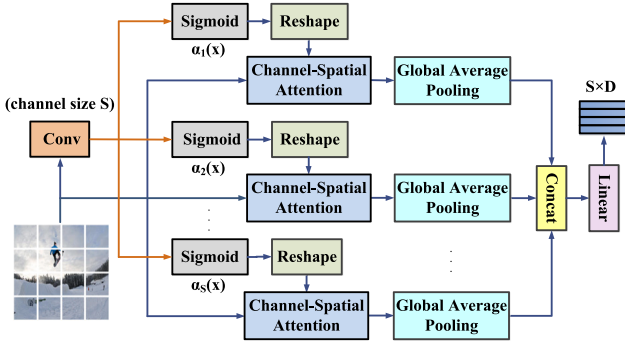


Fig. 2. Visual illustration of the Token Attention Module (TAM).

3. The approach

Fig. 1 shows the proposed Token-Reduction Transformer with Attribute Prediction (TRT-AP). It is an adaptive token attention framework, which can reduce ViTs' computation while maintaining the model's accuracy. The technical details are described in the following subsections.

3.1. Token reduction module

Given an image with $H \times W$ pixels, it is first divided into a sequence of patches $\{X_i \in \mathbb{R}^{h \times w}\}$, where $h \times w$ is the size of each patch. For clarity, suppose that a total of N patches are obtained in this way. Each patch X_i is embedded into a one-dimensional vector x_i , and these vectors are then assembled into a matrix X for convenience, where $X = [x_1, x_2, \dots, x_N]^T \in \mathbb{R}^{N \times M}$ and $M = h \times w$.

Instead of taking all tokens as the input for the Transformer, we develop a token attention module to learn a subset of important yet discriminative tokens. As mentioned previously, the motivation behind this treatment is to remove the redundant or even useless patches.

Fig. 2 illustrates the overall token attention module. Formally, given an image, the image will be first linearly converted into tokens, attention operations are then performed on these tokens. Let z_i be the i th token learned from the N original ones, $i = 1, 2, \dots, S$, with $S < N$. For clarity, let matrix Z collect the learned tokens, namely, $Z = [z_1, z_2, \dots, z_S]^T \in \mathbb{R}^{S \times D}$, where D is the dimensionality of the learned tokens.

Technically, the token attention module is mainly operated through the convolutional layers, channel-spatial attention mechanism and global average pooling. For the i th token z_i , it is obtained by the following equation:

$$z_i = D(\rho(X \odot \alpha_i(X))), \quad i = 1, 2, \dots, S, \quad (1)$$

where $\alpha_i(X) \in \mathbb{R}^{N \times (h \times w)}$ is a weighting tensor to be learned, $\odot$ is the Hadamard product, namely, the element-wise multiplication, $\rho(\cdot)$ is the spatial global average pooling that reduces the matrix $X \odot \alpha_i(X)$ to be a vector in $\mathbb{R}^N$, and D is a linear transformation that maps the resulting vector to $\mathbb{R}^D$ dimension. Thus, z_i represents a weight feature map achieved by scaling the input X with the weighting tensor.

In Eq. (1), each weighting tensor $\alpha_i(\cdot)$ is a weight map of size $N \times M$ conditioned on the input X . Technically, the weighting tensor is produced by a series of convolutional layers (channel-size S) followed by the activation of the sigmoid function. To keep the diversity and flexibility of the weight maps, the sigmoid function is applied to each of the S channel outputs of the convolution layer respectively. Accordingly, this structure can spatially learn the weighting tensor over the feature maps of the original patches. In image captioning tasks, objects and the relationships between them are important. Consequently, their

weighting values are higher compared to less significant regions, such as the background and noise.

As a result, totally S tokens are gathered to form the output tensor $Z = [z_i]_{i=1}^S \in \mathbb{R}^{S \times D}$. The overall process has a form of an element-wise spatial self-attention. Thus, the tokens are no longer the fixed splits of the input tensor, but a set of adaptive spatial selections.

3.2. Two-modal fusion Transformer encoder

As demonstrated in Fig. 1, the feature learning module comprises the Visual Transformer Encoder (VTE) for visual representation and the Attribute Transformer Encoder (ATE) for attribute representation. Architecturally, they are standard units of Transformers. Thus, two-modal representations can be learned. Different from the traditional framework, we propose to combine them by mapping the output of the VTE to the input of the ATE to enhance the performance of image captioning.

As shown in Fig. 1, an MLP layer is designed to classify and generate semantic attributes from the VTE:

$$z'_i = MLP(LN(z_i)), \quad i = 1, 2, \dots, S, \quad (2)$$

where LN represents Layer Normalization, which aims to normalize the input data's dimensions. MLP represents the Multi-Layer Perceptron. It is a fully connected feed-forward neural network, which is capable of handling complex non-linear relationships and learning more abstract feature representations.

Based on the output $[z'_i]_{i=1}^S$, a loss function incorporating attribute prediction is configured to guide model training through supervised learning. In this process, the number of attributes is set to be S . Then, these attributes are passed through a word embedding layer to produce S attribute tokens. The word embedding layer can help models better capture semantic relationships in a language by mapping words into a continuous vector space. For clarity, we denote them by $A = [a_i]_{i=1}^S \in \mathbb{R}^{S \times D}$.

Finally, the generated attention tokens Z and attribute tokens A are entered into the VTE and ATE respectively. It is noted that the VTE and the ATE both have L stacked Transformer blocks. As shown in Eqs. (3) and (4), in the l th Transformer block, the inter-generative attention tokens Z_l and attribute tokens A_l are first fed into the Multi-head Self-Attention (MSA), which allows the model to dynamically adjust its attention to information according to its relevance. The residual connection is also designed to solve the vanishing gradient problem during the training process, and to obtain better performances when the model goes deeper. Then, the generated Z'_l and A'_l are fed into the Feed-Forward Network (FFN), which can transform the encoded or decoded position vectors linearly, and strengthen the representation of position information and feature extraction. Likewise, residual connections have been added, the module finally generates outputs of the $(l+1)$ -th Transformer block Z_{l+1} and A_{l+1} respectively. Formally, for $l = 0, 1, \dots, L-1$, we have:

$$Z_0 = Z, Z'_l = MSA(Z_l) + Z_l, \quad Z_{l+1} = FFN(Z'_l) + Z'_l, \quad (3)$$

$$A_0 = A, A'_l = MSA(A_l) + A_l, \quad A_{l+1} = FFN(A'_l) + A'_l, \quad (4)$$

where Z_0 and A_0 represent the input of the first layer in the VTE and the ATE, respectively.

3.3. Transformer decoder

Similar to the encoder, the decoder also comprises L identical layers. The two-modal representations learned by the VTE and the ATE are fused together to guide the caption generation. Words are predicted sequentially. Specifically, let y_i denote the vector representation of the i th word, which is calculated as the sum of word embedding and

positional encoding. For the t th word, the decoder takes the sequence of all the previously-generated words as its input:

$$\mathbf{Y}_t = \{\mathbf{y}_0, \mathbf{y}_1, \dots, \mathbf{y}_{t-1}\} \subset \mathbb{R}^{d_w}, \quad (5)$$

where $\mathbf{y}_0$ is the start of the sentence, and d_w is the dimension of the word embedding. Different from the existing frameworks, our decoder learns to predict the next word conditioned on the fusion representations of visual and attribute tokens. In this process, the hidden state of the current step $\mathbf{h}_t$ depends on the feature embedding of the output token tensors and the words in history. That is, we have

$$\mathbf{h}_t^L = \text{decoder}(\mathbf{Z}_L \oplus \mathbf{A}_L, \mathbf{Y}_t), \quad (6)$$

where $\mathbf{Z}_L$ and $\mathbf{A}_L$ are the outputs of the last layer of Transformer encoder, $\oplus$ represents the two-modal fusion [19], $\mathbf{Y}_t$ is the word sequence feature of the previously generated sentence, and $\mathbf{h}_t^L$ is the output of the hidden unit in Transformer decoder that used to predict the current word $\mathbf{y}_t$.

The two-modal fusion approach [19] is different from the concatenation, element-wise product or sum, it uses the outer product to compute the multiplicative interaction between all elements of two vectors, which is more expressive than the above approaches. In addition, the multi-modal compact bilinear pooling (MCB) is proposed in this fusion approach, which can efficiently express the multi-modal features.

Finally, the output of the decoder is fed into the classifier over the vocabulary $\mathcal{V}$ to predict the next word, where $\mathcal{V}$ is a dictionary that generated by preprocessing the dataset. The conditional probability distribution of the words generated at time t is calculated by:

$$p(\mathbf{y}_t | \mathbf{Y}_t) = \text{softmax}(\mathbf{W}_y \mathbf{h}_t^L), \quad (7)$$

where $\mathbf{W}_y \in \mathbb{R}^{|\mathcal{V}| \times d_w}$ collects the parameters to be learned, and $|\mathcal{V}|$ is the number of words in the vocabulary.

3.4. Objective function and learning

Given the ground truth sequence $\mathbf{y}_{1:T}^*$, the objective is to minimize the following cross-entropy loss:

$$L(\theta) = - \sum_{t=1}^T \log(p_\theta(\mathbf{y}_t^* | \mathbf{y}_{1:t-1}^*)), \quad (8)$$

where θ collects the parameters in our model, and T is the maximum length of ground truth sentences in the training set.

Then, the non-differentiable CIDEr-D score is optimized by self-critical sequence training [20]:

$$\nabla_\theta L(\theta) = -\frac{1}{k} \sum_{i=1}^k (r(\mathbf{y}_{1:T}^i) - b) \nabla_\theta \log p_\theta(\mathbf{y}_{1:T}^i), \quad (9)$$

where k is the beam size, r is the CIDEr-D score function, and $b = (\sum_i r(\mathbf{y}_{1:T}^i))/k$ is the baseline, computed as the mean of the rewards obtained by the sampled sequences.

4. Experiments

4.1. Datasets

The experiments have been conducted first on the benchmark dataset MS COCO [21]. The dataset contains 123,287 images, every image is annotated with 5 different captions. By converting all sentences to lower case and filtering words that occur less than five times, totally we get a vocabulary of 10,096 words. For offline evaluation, the widely adopted ‘‘Karpathy’’ split [22] was followed, where 113,287, 5,000, and 5,000 images are used respectively for training, validation, and testing. In addition, the generated captions of the MS COCO official testing set were also uploaded for online evaluation.

To test the effectiveness of our approach, the performances on the NoCaps dataset [23] were also reported. It comprises 15,100 images

from Open Images, and covers more than 600 object categories. The images are divided into three domains: ‘in’, ‘near’ and ‘out’. In our comparative experiments, all the models were trained on the training split of the MS COCO dataset and tested on the validation split of NoCaps, which comprises images with novel objects and captions with unseen vocabularies.

4.2. Implementation details

Evaluation Metrics: The standard evaluation protocol is employed for the evaluation of approaches. Specifically, the captioning metrics include: BLEU@N, METEOR, ROUGE-L, CIDEr, and SPICE, which are denoted as B@N, M, R, C, and S for convenience.

Hyperparameters Settings: ViTs [9] are chosen as our baseline, due to its superior performance on the ImageNet. The input size is set to be 224×224 . The patch resolution is set to be 16×16 , and the number of tokens is thus produced as $N = 196$. The backbone comprises $L = 16$ blocks and $H = 8$ heads in each MSA layer, the inner-layer of the FFN has dimensionality of $d=2048$. Following the proposal given in Vaswani et al. [5], the dropout with the rate of 0.2 is employed on the output of each MSA and FFN sub-layer. In the Cross-Entropy training stage, our model was warmed up for 15 epochs with the learning rate linearly increased to 1×10^{-4} . Then the learning rate was set to be 2×10^{-5} in next 15 epochs. The batch size was set to 256. After the 30-epoch Cross-Entropy training stage, our model was optimized with CIDEr reward with 5×10^{-6} learning rate for 30 epochs. The ADAM optimizer were used in both stages, and the beam size was set as 5. The generation ends when the ‘EOS’ token is predicted, or when the maximum length of 20 words is reached, the output dimension of the word embedding d_w was set as 512.

Training Settings: For the training of the whole network, first of all, the process of embedding images into tokens is pre-processed. Secondly, the embedded tokens are sent to the token attention module to learn informative tokens. Thirdly, the learned tokens go through two-modal fusion Transformers under the encoder–decoder framework, and finally generate captions. It is worth noting that the entire training process is end-to-end.

4.3. Performance comparison

4.3.1. Offline evaluation

As shown in Table 1, our approach was compared with several recent models for image captioning. The compared models include attention-based models: Up–Down [11], AoANet [24], VPNet [12]. Transformer-based models: M² Transformer [25], X-LAN [26], CBTIC [14], GRIT [15]. The current more popular vision-language pre-training (VLP) models: LEMON [27], OFA [28]. All approaches were evaluated on the MS COCO ‘‘Karpathy’’ test split, using both single model and ensemble configurations.

Single Model. In the upper panel of Table 1, the performances of our approach and the state-of-the-art models are reported. For fair comparison, all the models were first trained under cross-entropy loss, then optimized for CIDEr-D score. It can be seen that our approach achieves the competitive performances in most metrics. For example, compared with the attention-based model, our TRT-AP has advantage over the Up–Down [11], AoANet [24] and VPNet [12] in all metrics. In addition, compared with the Transformer-based model, our approach outperforms most other models, but performs slightly worse than GRIT [15], which takes full advantage of Grid and Region features for the captioning task. Moreover, compared with LEMON [27] and OFA [28], which focus on studying VLP, there is still a gap. That is largely due to the fact that these two models are actually large models, which require large-scale data to pre-train and have larger calculations than our TRT-AP relatively. In general, our proposed model is competitive.

Table 1Performances of our model and other state-of-the-art models on the MS COCO “Karpathy” test split. Σ indicates ensemble or fusion models.

Model	Cross-entropy loss						CIDEr-D optimization					
	B@1	B@4	M	R	C	S	B@1	B@4	M	R	C	S
Single Model												
Up-Down [11]	77.2	36.2	27.0	56.4	113.5	20.3	79.8	36.3	27.7	56.9	120.1	21.4
AoANet [24]	77.4	37.2	28.4	57.5	119.8	21.3	80.2	38.9	29.2	58.8	129.8	22.4
M ² Transformer [25]	–	–	–	–	–	–	80.8	39.1	29.2	58.6	131.2	22.6
X-LAN [26]	78.0	38.2	28.8	58.0	122.0	21.9	80.8	39.5	29.5	59.2	132.0	23.4
CBTIC [14]	78.0	37.5	29.1	58.2	122.6	22.3	81.4	39.6	29.9	59.1	136.9	24.0
VPNet [12]	78.6	38.2	28.6	58.0	121.0	21.9	80.9	39.7	29.3	59.2	130.4	23.2
GRIT [15]	–	–	–	–	–	–	83.5	41.9	30.5	60.5	142.2	24.2
LEMON [27]	–	41.5	30.8	–	139.1	24.1	–	42.6	31.4	–	145.5	25.5
OFA [28]	–	43.9	31.8	–	145.3	24.8	–	44.9	32.5	–	154.9	26.6
TRT-AP	78.8	40.3	29.8	58.3	131.3	22.1	82.4	40.1	29.9	60.5	139.2	24.1
Ensemble/Fusion												
GCN-LSTM Σ [29]	77.4	37.1	28.1	57.2	117.1	21.1	80.9	38.3	28.6	58.5	128.7	22.1
AoANet Σ [24]	78.7	38.1	28.5	58.2	122.7	21.7	81.6	40.2	29.3	59.4	132.0	22.8
M ² Transformer [25]	–	–	–	–	–	–	82.0	40.5	29.7	59.5	134.5	23.5
X-LAN Σ [26]	78.8	39.1	29.1	58.5	124.5	22.2	81.6	40.3	29.8	59.6	133.7	23.6
VPNet Σ [12]	79.3	39.1	29.1	58.7	124.4	22.2	81.8	40.9	29.8	59.7	133.9	23.6
CBTIC Σ [14]	79.1	39.4	29.8	59.3	127.8	23.1	82.7	41.1	30.4	60.1	140.3	24.3
TRT-AP Σ	81.0	40.7	30.2	60.1	135.3	23.3	83.2	41.5	30.5	61.1	142.1	24.7

Table 2

The consume time of our model and other state-of-the-art models on the MS COCO “Karpathy” test split (single model).

Model	Up-Down	AoANet	M ² Transformer	X-LAN	CBTIC
Time (hours)	9.0	13.2	11.1	14.9	12.5
Model	VPNet	GRIT	LEMON	OFA	Our
Time (hours)	10.9	8.6	28.7	26.5	6.7

Ensemble Model. By following the commonly-used proposal on model ensemble given in Huang et al. [24], the output probability distributions of multiple and independently trained instances are averaged. The bottom panel of Table 1 reports the performances of all ensemble models. More specifically, ensembles of four models are used, which are trained from different random seeds. As we can see from the results, our approach achieves the best score in most of metrics. Especially, compared with the comprehensive model CBTIC [14], our approach improves 7.5 CIDEr points under the Cross-Entropy loss, and improves 1.8 CIDEr points under the CIDEr optimization.

In Table 2, the paper compares the consume time of our approach with other state-of-the-art algorithms on the MS COCO “Karpathy” test split. It can be seen that compared with other models, our approach is more time-saving, especially for large models [27,28], and the time spent is only one-fifth of them.

4.3.2. Online evaluation

The performances of our approach and other state-of-the-art algorithms were also reported on the online MS COCO test server, for which ground-truth annotations are not publicly available. The results were reported in Table 3, they show the performance leader-board with 5 reference captions (c5) and 40 reference captions (c40). For fair comparison, all models used an ensemble configuration and trained the model on the “Karpathy” training split. From the results shown in the Table 3, our work surpasses all the state-of-the-art captioning approaches. Specifically, our approach achieves the high score of 139.3 on the most important CIDEr-D (c40) metric, achieving an advancement of 1.2 CIDEr points with respect to the best model CBTIC [14].

4.4. Results on the NoCaps dataset

Table 4 shows the results of our approach and other state-of-the-art algorithms on the NoCaps dataset, including models: NBT [23], Up-Down [11], M² Transformer [25], Oscar [32], VIVO [33], Enc-Dec [34]. As shown in the table, TRT-AP has achieved remarkable

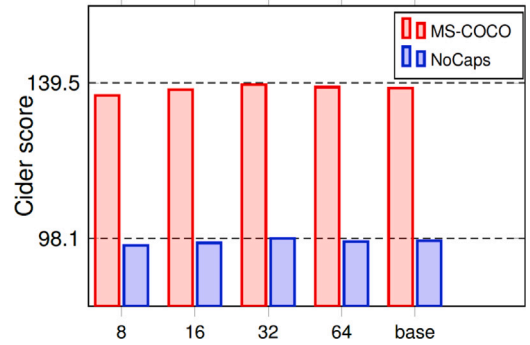


Fig. 3. The accuracy of our model on the MS-COCO and NoCaps datasets under different number of tokens within the token attention module.

improvement, especially for in-domain and near-domain images. For example, compared with Enc-Dec [34], the CIDEr score is improved by 8.8 for the in-domain part, 7.5 for the near-domain part, and 0.4 for the out-of-domain part. Although Oscar [32], VIVO [33], Enc-Dec [34] are VLP models and use large-scale data for pre-training, these models have more or less shortcomings, and our approach is superior to them in the use of strategies. This evidences that token attention and attribute prediction improves the model’s ability to recognize a wide range of visual objects.

4.5. Ablation study

To evaluate the validness of the proposed token attention, various number of tokens with $S=8, 16, 32, 64$ are tried. The token attention module learns to attend an image into 8, 16, 32, 64 tokens. The channel-spatial attention function α_i is implemented with four 3×3 convolutional layers, whose channel size is identical to the number of tokens. As shown in Table 5, we report the amount of computation (GFLOPs) and the number of parameters of each token attention module. They depend on the input size and the hyper parameter setting. It is seen that computation reduction is significantly achieved with smaller number of tokens, and the parameters of token attention module increase as the number of tokens increases.

The effect of different number of tokens using our token attention module is then considered on the MS COCO and NoCaps datasets, even includes the baseline (without the token attention), the baseline ViTs

Table 3

Performances of our model and other state-of-the-art models on the online MS-COCO test server.

Model	B@1		B@4		M		R		C	
	c5	c40	c5	c40	c5	c40	c5	c40	c5	c40
Up-Down [11]	80.2	95.2	36.9	68.5	27.6	36.7	57.1	72.4	117.9	120.5
T-MaC [10]	80.3	95.4	38.1	70.6	28.7	37.9	58.1	73.0	125.5	128.3
AoANet [24]	81.0	95.0	39.4	71.2	29.1	38.5	58.9	74.5	126.9	129.6
VPNet [12]	81.0	95.3	39.6	71.6	29.2	38.7	59.1	74.6	126.6	128.9
M ² Transformer [25]	81.6	96.0	39.7	72.8	29.4	39.0	59.2	74.8	129.3	132.1
X-LAN [26]	81.4	95.7	40.0	72.4	29.7	39.3	59.5	75.2	130.2	132.8
RSTNet [30]	82.1	96.4	40.0	73.1	29.6	39.1	59.5	74.6	131.9	134.0
DLCT [31]	82.4	96.6	40.6	74.0	29.8	39.6	59.8	75.3	133.3	135.4
CBTIC [14]	82.3	96.5	40.5	73.5	30.0	39.6	59.6	75.0	135.0	138.6
TRT-AP	82.7	96.8	41.0	74.5	30.0	40.3	60.6	75.5	135.5	139.3



GT: An orange cat rubbing up against a black bag.
Transformer: Two cats of different colors standing next to a black bag.
TRT-AP: An orange cat and a gray cat standing next to a black bag on a stairs.



GT: A red stop light on a pole on the corner of a road.
Transformer: A traffic light sitting on the side of a road.
TRT-AP: A red traffic light on the corner of a road.



GT: A kitchen with a window that shows into the bathroom.
Transformer: A wood kitchen with a window to an adjacent bathroom.
TRT-AP: A clean kitchen with a small set of burners and a tea kettle on the counter top of a neat.



GT: A number of men and women posing for a picture.
Transformer: A group of people standing next to a TV.
TRT-AP: A number of men and women poses for a photo.

Fig. 4. Examples of image captioning results by Transformer and our proposed TRT-AP, coupled with ground truth sentences.**Table 4**

Performances of our model and other state-of-the-art models on the NoCaps validation split sets for different domains.

Model	In-domain		Near-domain		Out-domain		Overall	
	C	S	C	S	C	S	C	S
NBT [23]	70.9	11.1	58.7	10.0	35.0	6.8	52.4	9.1
Up-Down [11]	74.3	11.5	56.9	10.3	30.1	8.1	54.3	10.1
M ² Transformer [25]	85.7	12.1	–	–	38.9	8.9	64.5	11.1
Oscar [32]	79.6	12.3	66.1	11.5	45.3	9.7	63.8	11.2
VIVO [33]	88.8	12.9	83.2	12.6	71.1	10.6	81.5	12.2
Enc-Dec [34]	92.6	12.5	88.3	12.1	94.5	11.9	90.2	12.1
TRT-AP	101.4	12.8	95.8	12.5	94.9	12.1	95.1	12.8

Table 5

The amount of computation (GFLOPs) and number of parameters of each token attention module under different number of tokens.

Number	8	16	32	64	base
GFLOPs	2.5	3.3	3.9	4.5	8.6
params (M)	1.11	1.87	2.79	3.56	6.91

use 196 tokens. As can be seen from the bar chart in Fig. 3, there is a consistent increase in accuracy until $S=32$. It is noted that the best CIDEr score is achieved with $S=32$, and the amount of computation is appropriate at the same time. When the reduction of GFLOPs is more than 40%, the accuracy even improves. When using the baseline, it performs slightly worse than our token attention mechanism. This fact indicates that a large number of tokens are actually redundant. The final results imply that our approach can effectively identify tokens which are useful for the final prediction.

4.6. Qualitative analysis

Fig. 4 shows some examples of the captions generated by our proposed TRT-AP and the Transformer, coupled with the human-annotated ground truth (GT) sentences. Compared with the captions generated by the Transformer, TRT-AP produces more accurate and rich descriptive sentences, bringing more details in attributes. For example, in the first picture, TRT-AP can accurately describe the color of two dogs, “orange” and “black”, whereas the Transformer cannot do this. Similarly, in the second picture, TRT-AP not only recognizes traffic lights at intersections, but also recognizes light colors. But the Transformer fails to do this job. In the third picture, although both Transformer and our approach can identify large objects such as kitchens and Windows, only TRT-AP can identify small objects such as burners and tea kettles. In the last picture, TRT-AP not only describes the large number of people in the picture, but also identifies the gender of the people. On the contrary, the Transformer fails to do this. In summary, the generated captions of our model is more complete and detailed, which further demonstrates the effectiveness of our approach.

5. Conclusions

In this work, we have proposed a novel Transformer model with token reduction and attribute prediction for image captioning. By performing attention operations on tokens generated by ViTs, this model removes unnecessary and redundant data, and generates tokens that are more effective and important for captioning tasks. At the same time, the MLP classifier is used to predict the attributes of images. Accordingly, both of the learned attributes and tokens are entered into the two-modal Transformer encoder-decoder, which can strengthen the relationship between the vision representations and language representations. Thus, it is helpful to generate the captions related to

images. The experimental results demonstrate that our proposed model can generate captions correctly. The performance comparisons with the existing state-of-the-art models also show the effectiveness and competitiveness of our model.

Nevertheless, our work still has some limitations. For example, as described in Section 3.3, how to effectively combine visual features and attribute features is not explored in this paper, which will affect the final result. In addition, there are still gaps in our work compared to current popular VLP models.

In the future work, we will focus on addressing the limitations of our model. First, we would like to explore how to combine visual and attribute features, so that the two features can perfectly interact. Second, a reasonable combination of region features and token features is also an effective way to strengthen the visual features. Finally, the use of large models can also effectively improve the captioning task.

CRedit authorship contribution statement

Lifei Song: Writing – original draft, Validation, Software, Data curation. **Ying Wang:** Visualization, Methodology, Funding acquisition. **Linsu Shi:** Project administration, Formal analysis. **Jiazhong Yu:** Supervision, Investigation, Funding acquisition. **Fei Li:** Investigation, Formal analysis, Conceptualization. **Shiming Xiang:** Writing – review & editing, Writing – original draft, Validation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This research was supported by the National Key Research and Development Program of China under Grant No. 2018AAA0100400, and the National Natural Science Foundation of China under Grant 62076242.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.patrec.2024.11.022>.

Data availability

The data that has been used is confidential.

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